

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image segmentation tooling for **95 brain subparts**, handling preprocessing, orientation correction, model inference, generated masks, and large-volume MRI data paths.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Debugged model-output pipeline issues across conformation, orientation, label mapping, PyTorch/ONNX-style inference, and binary mask generation in a mature open-source codebase.

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across workflow preparation.
- Built a resilient automation backend with retries, session recovery, public-site API lookups, and state validation, reducing manual filing time by **85%**.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Open Source

[Contributions](#) 

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review processes for reliability, security, and integration quality.

Projects

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, Vector Memory, JSONL*

- Built a multi-agent research workflow with analyst roles, bull/bear debate, trader, risk review, portfolio approval, typed Pydantic state, SQLite checkpoints, decision memory, and JSONL traces.
- Added **vector memory** for semantically similar decision lessons, validation harnesses, reward loops, and backtests against multiple baselines so agent behavior and retrieval quality could be inspected.

Swish Support System | *Python, FastAPI, LLM APIs, PostgreSQL, Redis, Langfuse*

GitHub 

- Built an **AI-assisted support backend** where LLM calls classify messy customer complaints, intent, severity, ambiguity, and photo-evidence usefulness while deterministic code controls policy actions.
- Maintained conversation context, evidence artifacts, order/customer state, traceable LLM assessments, and regression tests for multi-turn support workflows.

Omni-SSM Voice Agent | *Python, PyTorch, ONNX Runtime, Sherpa-ONNX, Moonshine, Hindi ASR, Qwen TTS, Piper*

- Built an on-device STT/TTS research loop with **SSAMBA streaming CTC**, **Moonshine Hindi ASR tuning** for a non-English speech dataset, Piper low-latency TTS, Qwen expressive TTS experiments, and WebRTC barge-in benchmarks.
- Implemented manifest-driven data loading, teacher/student distillation and partial fine-tuning scripts, LoRA SFT data prep, multilingual TTS intelligibility/prosody gates, and latency reports for STT, TTS, first-audio, and device-memory constraints.

Technical Skills

Languages: Python, Go, TypeScript, C++

Frameworks: FastAPI, Agno, PyTorch, ONNX Runtime, Sherpa-ONNX, Node.js

Tools: PostgreSQL, SQLite, Redis, Vector Memory, Docker, GitHub Actions, GCP, Langfuse, Colab, Prometheus